Test 9 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature (Questions 1–15)

1. **Read the passage: *"Sara observed the museum exhibits, noting the artifacts from the Bronze Age and taking photographs for her project."***
   Question: What was Sara mainly doing at the museum?
2. **List two types of artifacts mentioned in the passage.**
3. **Rewrite the sentence in past perfect tense: *"Sara takes notes in her notebook."***
4. **Correct the grammar: *"Neither of the boys have completed the assignment."***
5. **Identify the adjective in: *"The ancient manuscripts fascinated the students."***
6. **Write a short story (80–100 words) about discovering a hidden library.**
7. **Compose a formal letter to the museum requesting permission to study an exhibit.**
8. **Transform into passive voice: *"Sara photographed the artifacts."***
9. **Define *simile* and use it in an original sentence.**
10. **Summarize the passage in 40–50 words.**
11. **Turn into a question: *"The students completed the history project on time."***
12. **Write a paragraph describing your favorite book (60–80 words).**
13. **Identify the main idea of this passage: *"Technology has transformed communication across the globe."***
14. **Correct the punctuation: *"Its raining heavily; we must take umbrella"***
15. **Explain the difference between first-person and third-person narration with examples.**

Section B: Mathematics (Questions 16–35)

16. **Solve: $4x - 9 = 15$**
17. **Factorize: $x^2 + 9x + 20$**
18. **Simplify: $(6a^2b^3)(2ab^2)$**
19. **Solve the system:**
    $2x + 3y = 12$
    $4x - y = 5$
20. **Calculate the circumference of a circle with radius 10 cm.**
21. **Solve: $7(3x - 2) = 2x + 26$**
22. **Expand: $(x - 4)(x + 7)$**
23. **Find the HCF of 36 and 60.**

24. A car travels 540 km in 9 hours. Find its average speed.
25. Solve: $6y/4 - 5 = 7$
26. Find the roots of $x^2 - 5x - 14 = 0$.
27. Find the area of a square with diagonal 14 cm.
28. Simplify: $(5a^3b^2) \div (ab)$
29. A bag contains 4 red, 7 blue, and 9 green balls. Probability of picking a green ball?
30. Solve: $3(x - 6) + 4 = 2x + 8$
31. Convert 0.875 into a fraction.
32. A polygon has 10 sides. Find the sum of interior angles.
33. Solve: $9(x - 4) = 3x + 18$
34. The ratio of boys to girls in a class is 7:5. If there are 35 boys, how many girls?
35. Simplify: $(8x^2y^3)(3xy^2)$

Section C: Physics (Questions 36–50)

36. State Newton's Second Law of Motion.
37. A 15 N force is applied to a 3 kg object. Find its acceleration.
38. Define speed and velocity with examples.
39. A car accelerates from 20 m/s to 50 m/s in 10 seconds. Find the acceleration.
40. Define potential energy and give an example.
41. A 5 kg object is lifted to 10 m. Calculate its potential energy ($g = 10\ m/s^2$).
42. Explain conservation of momentum with an example.
43. Define power and give its SI unit.
44. A force of 50 N moves a box 6 m. Calculate the work done.
45. Differentiate between scalar and vector quantities.
46. State the law of reflection.
47. Give an example of energy transfer by conduction.
48. A 12 Ω resistor has 4 A current. Find the voltage.
49. Explain Ohm's Law.
50. Calculate the energy consumed by a 200 W heater used for 3 hours.

Section D: Chemistry (Questions 51–65)

51. Define an acid and give an example.
52. Write the formula for sodium hydroxide.
53. Balance: $Fe + O_2 \rightarrow Fe_2O_3$
54. Differentiate between strong and weak acids.
55. Give one example each of homogeneous and heterogeneous mixtures.
56. Write electron configuration of oxygen.

57. **Define oxidation with an example.**
58. **Write products of metal + acid reaction.**
59. **Is melting ice a chemical or physical change?**
60. **Describe Mendeleev's contribution to the periodic table.**
61. **Give the chemical formula for sulfuric acid.**
62. **Explain isotope with an example.**
63. **Differentiate between metals and nonmetals.**
64. **Write the reaction of combustion of ethane.**
65. **Explain the pH value of neutral, acidic, and basic solutions.**

Section E: Biology (Questions 66–80)

66. **Define photosynthesis and write its chemical equation.**
67. **Role of mitochondria in cells.**
68. **Compare plant and animal cells.**
69. **Describe the structure of human heart.**
70. **Define diffusion and give an example.**
71. **Role of liver in digestion.**
72. **Two bacterial diseases affecting humans.**
73. **Define habitat.**
74. **Difference between vertebrates and invertebrates.**
75. **Stages of human digestive process.**
76. **Types of joints in human body.**
77. **Function of white blood cells.**
78. **Explain nitrogen cycle.**
79. **Role of roots in plants.**
80. **Explain transpiration in plants.**

Section F: Social Studies / History / Geography (Questions 81–90)

81. **Define democracy.**
82. **Three longest rivers in Asia.**
83. **Causes of Industrial Revolution.**
84. **Define greenhouse effect.**
85. **Capital of Australia.**
86. **Explain rule of law.**
87. **Identify major mountain ranges in South America.**
88. **Effects of soil erosion.**

89. Name three early civilizations in the Indus Valley.
90. Difference between latitude and longitude.

Section G: Computer / ICT (Questions 91–95)

91. Define programming language.
92. Give two examples of output devices.
93. Functions of database management system (DBMS).
94. Difference between RAM and ROM.
95. Explain purpose of antivirus software.

Section H: Islamiyat / Religious Studies (Questions 96–100)

96. List the five pillars of Islam.
97. Explain the significance of Salah (prayer) in Islam.
98. Define Zakat and its role in society.
99. Describe the life of Prophet Ibrahim (AS) and key lesson.
100. Explain the story of Prophet Yusuf (Joseph) and the moral lesson.